

Day 5 — Why It Works — The Algebra

Module: Subtraction — Nikhilaṃ (All from 9) · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: derive the place-value identity that makes the rule work.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *Nikhilaṃ Navataścaramaṃ Daśataḥ* (IAST: Nikhilaṃ Navataścaramaṃ Daśataḥ; lit. “all from 9, the last from 10”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **State the identity.** For N with k digits, $10^k - N$ can be computed as $((10^k - 1) - N) + 1$.
2. **Why “all from 9 last from 10” works.**
 - $(10^k - 1)$ is a number with k nines: 9, 99, 999, 9999, ...
 - $(10^k - 1) - N$: each digit of N is subtracted from 9. The result has no borrows because each digit of $(10^k - 1)$ is 9.
 - Adding 1 to that bumps the last digit by 1, which is equivalent to “the last digit from 10” rather than “from 9.”
3. **Worked algebraic example.** $N = 437$, $k = 3$.
 - $999 - 437 = 562$ (digit-wise: 9-4, 9-3, 9-7)
 - $562 + 1 = 563 = 1000 - 437$. ✓
4. **The general formula.** $10^k - N = (10^k - 1) - N + 1$ — a foundational identity in modular arithmetic and binary computer arithmetic.
5. **Senior bridge.** This is the same identity that underlies *two’s complement* in binary computer arithmetic — tomorrow’s topic.

Activity (15 min)

A scaled version of *Complement Flashcards* (see [activities/](#)). Today's slice: focus on the part of the activity that reinforces Why It Works — The Algebra.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See `homework.md` for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).